EXPERIMENT-32

32. C Program to Demonstrate the Difference Between DSA and RSA Digital Signatures

AIM

To write a C program to demonstrate that DSA generates different digital signatures for the same message using different random values of k, whereas deterministic RSA generates the same signature for the same message and private key.

PROCEDURE

1. Start the program.
2. Initialize the message hash and private key.
3. Select the DSA parameters p, q, and g.
4. Generate the first random value k1.
5. Calculate the first DSA signature `(r1, s1)`.
6. Generate another random value k2.
7. Calculate the second DSA signature `(r2, s2)`.
8. Compare both DSA signatures.
9. Demonstrate deterministic RSA signing using the same message and private key.
10. Display the signature differences.
11. Stop the program.

C PROGRAM

```
#include <stdio.h>

#define P 23
#define Q 11
#define G 2
```

```
int mod_pow(int a, int b, int mod)
{
    int result = 1;

    while (b > 0) {
        result = (result * a) % mod;
        b--;
    }

    return result;
}

int mod_inverse(int a, int mod)
{
    int i;

    for (i = 1; i < mod; i++) {
        if ((a * i) % mod == 1)
            return i;
    }

    return -1;
}

void dsa_sign(int h, int x, int k,
              int *r, int *s)
{
    int kinv;

    *r = mod_pow(G, k, P) % Q;
    kinv = mod_inverse(k, Q);

    *s = (kinv * (h + x * (*r))) % Q;
}

int main()
{
    int h = 5;
    int x = 3;

    int k1 = 2, k2 = 4;
```

```
    int r1, s1, r2, s2;

    printf("DSA AND RSA SIGNATURE COMPARISON\n");

    dsa_sign(h, x, k1, &r1, &s1);
    dsa_sign(h, x, k2, &r2, &s2);

    printf("\nFirst DSA Signature:");
    printf("\nr = %d, s = %d", r1, s1);

    printf("\n\nSecond DSA Signature:");
    printf("\nr = %d, s = %d", r2, s2);

    printf("\n\nDSA Result:");
    if (r1 != r2 || s1 != s2)
        printf("\nSignatures are DIFFERENT.");

    printf("\n\nRSA Result:");
    printf("\nSame message and private key");
    printf("\nDeterministic RSA signatures are SAME.");

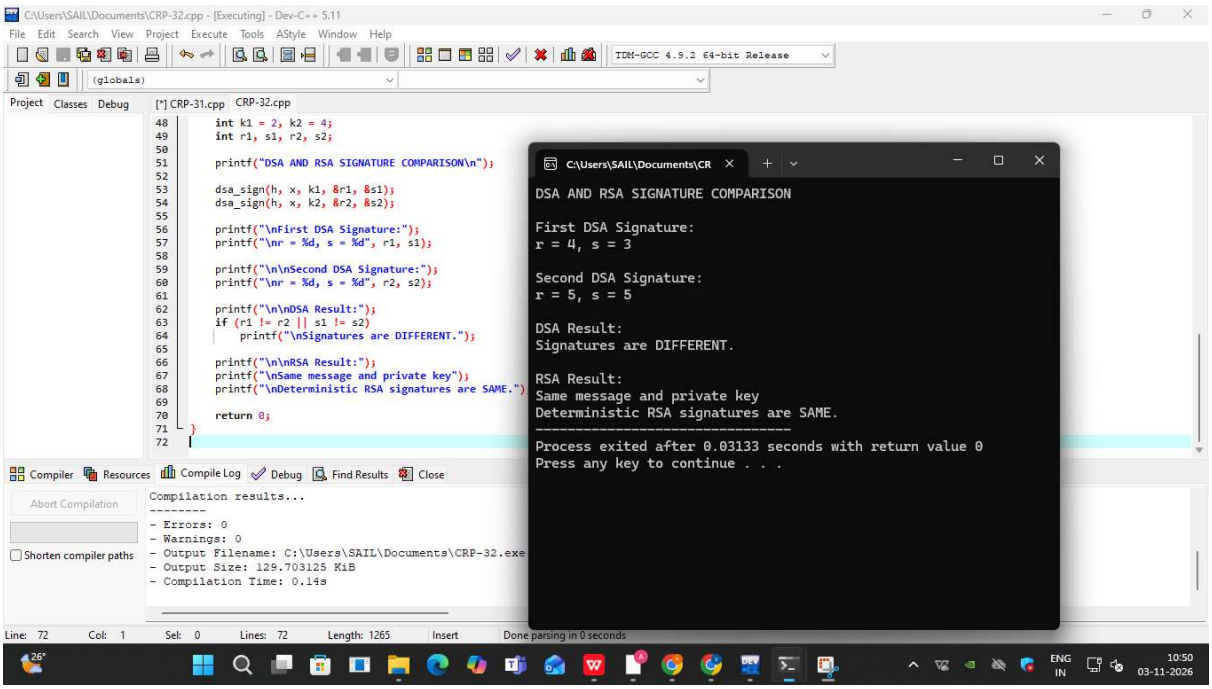
    return 0;
}
```

SAMPLE INPUT

```
Message hash = 5
Private key = 3

First random k = 2
Second random k = 4
```

SAMPLE OUTPUT



EXPLANATION

1. DSA (Digital Signature Algorithm)

- DSA uses a random number k for every signature.
- Different k values produce different signatures.
- The same message can have different signatures.
- The random value k must be kept secret and must never be reused.

2. RSA (Rivest–Shamir–Adleman)

- Traditional deterministic RSA signatures produce the same signature for the same message and private key.
- RSA-PSS uses random salt, so its signatures can differ.

RESULT

Thus, the C program successfully demonstrates the difference between DSA and RSA digital signatures. DSA produces different signatures for the same message when different random values of k are used, whereas deterministic RSA produces the same signature for the same message and private key.